Experiment - 07

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Q1.
Consider a relation R having attributes as R(ABCD), functional dependencies are given below:

AB->C, C->D, D->A

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution: Given,
AB → C, C → D, D → A

B never appears on RHS so B must be a part of candidate keys

Finding closures,
B+ = B
AB+ = ABCD
BC+ = BCDA
BD+ = BDAC

Candidate Keys: {AB, BC, BD}
Prime Attributes: {A, B, C, D}
Non-Prime Attributes: {}

Checking FDs for BCNF,
AB → C (AB is a super key)
C → D (C is not a super key so it violates BCNF property)

And since all attributes are prime attributes, relation is in 3NF.

Q2. Relation R(ABCDE) having functional dependencies as:
A->D, B->A, BC->D, AC->BE

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution: Given,
A → D
B → A
BC → D
AC → BE

C never
appears
on
RHS so
C must
be a
part of
the
candida
te key

Finding closures: C+
= C
AC+ = ACBED
BC+ = BCADE
CD+ = CD
CE+ = CE

Candidate Keys: {AC, BC}
Prime Attributes: {A, B, C}
Non-Prime Attributes {D, E}

Checking FDs for BCNF,
A → D (A is not a super key, hence violates property for BCNF)

Checking FDs for 3NF,
A → D (Neither A is super key nor is D a prime attribute, hence violates 3NF)

Checking FDs for 2NF,
A → D (A is a proper subset of candidate key AC and D is a non-prime attribute, hence violates 2NF)

So, it is in 1NF.

Q3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:

B->A, A->C, BC->D, AC->BE

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution: Given,
B → A
A → C
BC → D
AC → BE

A+ = ACBED
B+ = BACED

Candidate Keys: {A, B}
Prime Attributes: {A, B}
Non-Prime Attributes: {C, D, E}

Checking FDs for BCNF,
B → A (B is a super key)
A → C (A is a super key)
BC → D (BC is a super key)
AC → BE (AC is a super key)

So the table is in BCNF.

Q4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:

A->BCD, BC->DE, B->D, D->A

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:
Given,
A → BCD
BC → DE
B → D
D → A

F never appears on RHS, so F must be a part of the candidate key

Finding closures,
F+ = F
AF+ = ABCDEF
BF+ = BFDACE
CF+ = CF
DF+ = DFABCE
EF+ = EF

Candidate Keys: {AF, BF, DF}
Prime Attributes: {A, B, D, F}
Non-Prime Attributes: {C, E}

Checking FDs for BCNF,
A → BCD (A is not a super key)

Checking FDs for 3NF,
A → BCD (A is not a super key and neither are C is non-prime)

Checking FDs for 2NF,
A → BCD (A is a subset of candidate key AF)

So highest NF is 1NF.

Q5. Relation R(ABCDE) having dependencies as:

CE -> D

D ->B

C -> A

Highest normal form? C.K set?

Solution:
Given, CE
→ D
D → B
C → A

C and E never appear on RHS so CE must be a part of candidate key

Finding closures,

CE+ = CEDBA

Candidate Keys: {CE}
Prime Attributes: {C, E}
Non-Prime Attributes: {A, B, D}

Checking BCNF,
CE → D (CE is a super key)
D → B (D is not a super key)

Checking 2NF
C → A (C is a proper subset of candidate key CE, and A is non-prime, hence violates 2NF)

Highest Normal Form: 1NF

Q6. Relation R(ABCDEF) having dependencies as:

AB -> C

DC -> AE

E -> F

Highest normal form? C.K set?

Solution:
Given, AB
→ C
DC → AE
E → F

B and D never appear on RHS, so both B and D must be part of every candidate key.

Finding closures,

BD+ = BD
ABD+ = ABCDEF
BCD+ = ABCDEF

Candidate Keys: {ABD, BCD}
Prime Attributes: {A, B, C, D}
Non-Prime Attributes: {E, F}

Checking BCNF,
AB → C (AB is not a super key)

Checking 3NF,
DC → AE (DC is not a super key and E is non-prime)

Checking 2NF,
DC → E (DC subset of candidate key BCD)

Highest NF: 1NF

Q7. Relation R(ABCDEFGHI) having dependencies as:

AB -> C

BD -> EF

AD -> GH

A -> I

Highest normal form? C.K set?

Solution: Given,
AB → C
BD → EF
AD → GH
A → I

A, B, D never appear on RHS, so A, B and D must be part of every candidate key
ABD+ = ABCDEFGHI

Candidate Keys: {ABD}
Prime Attributes: {A, B, D}
Non-Prime Attributes: {C, E, F, G, H, I}

Checking BCNF,
AB → C (AB is not a super key)

Checking 3NF,
AB → C (AB is not a super key and C is non-prime)

Checking 2NF,
AB → C (AB is a proper subset of candidate key ABD and C is non-prime)

Highest NF: 1NF

Q8. Relation R(ABCDE) having dependencies as:

AB -> CD

D -> A

BC -> DE

Highest normal form? C.K set?

Solution: Given,

AB → CD
D → A
BC → DE

Since B never appears on RHS, B must be part of every candidate key.

Finding closures,

B+ = B
AB+ = ABCDE
BD+ = ABCDE
BC+ = ABCDE

Candidate Keys: {AB, BD, BC}
Prime Attributes: {A, B, C, D}
Non-Prime Attributes: {E}

Checking BCNF,
D → A (D is not a super key)

Checking 3NF,
D → A (D is not a super key, but A is prime)
Highest NF: 3NF

Q9. Relation R(ABCDE) having dependencies as:
BC -> ADE
D -> B
Highest normal form? C.K set?

Solution: Given,
BC → ADE
D → B

C, and E never appear on RHS, so C and E must be part of every candidate key.

Finding closures,
C+ = C

BC+ = ABCDE
CD+ = ABCDE

Candidate Keys: {BC, CD}
Prime Attributes: {B, C, D}
Non-Prime Attributes: {A, E}

Checking BCNF,
D → B (D is not a super key)

Checking 3NF,
D → B (D is not a super key, but B is prime)
BC → ADE (BC is a super key)

Highest NF: 3NF

Q10. Relation R(ABC) having dependencies as:
A -> B
B -> C C -> A
Highest normal form? C.K set?

Solution:
Given,
A → B
B → C
C → A

Finding closures,
A+ = ABC
B+ = ABC
C+ = ABC

Candidate Keys: {A, B, C}
Prime Attributes: {A, B, C}
Non-Prime Attributes: {}

All are determinants are super keys so BCNF

Highest NF: BCNF

Q11. Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

F1 -> F3
F2 -> F4
(F1,F2) -> F5

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Solution: Given,
F1 → F3
F2 → F4 (F1,
F2) → F5

F1 and F2 never appear on RHS, so they must be part of every candidate key.

Finding closures,
F1F2+ = F1F2F3F4F5

Candidate Key: {F1F2}
Prime Attributes: {F1, F2}
Non-Prime Attributes: {F3, F4, F5}

Checking BCNF,
F1 → F3 (F1 is not a super key)

Checking 3NF,
F1 → F3 (F1 is not a super key and F3 is non-prime)

Checking 2NF,
F1 → F3 (F1 is a proper subset of candidate key F1F2 and F3 is non-prime)

Highest NF: 1NF

Q12. Consider a relation R(ABCDEFGHIJ) having FDs as:

AB -> C

AD -> GH

BD -> EF

A -> I

H -> J

Identify that relation is in which normal form and the candidate key by determining the prime and non-prime attributes.
Apply the decomposition if needed to remove the redundancy.
Identify how many new tables will be formed after the decomposition.

Solution:
Given, AB
→ C
AD → GH
BD → EF
A → I
H → J

A, B and D never appear on RHS, so A, B and D must be part of every candidate key

Finding closures,
ABD+ = ABCDEFGHIJ

Candidate Key: {ABD}
Prime Attributes: {A, B, D}
Non-Prime Attributes: {C, E, F, G, H, I, J}

Checking BCNF,
AB → C (AB is not a super key, so relation violates BCNF)

Checking 3NF,
AB → C (AB is not a super key and C is non-prime, so violates 3NF)

Checking 2NF,
AB → C (AB is a proper subset of candidate key ABD and C is non-prime, so it violates 2NF)

Highest NF: 1NF

Decomposition to remove redundancy

Since the relation violates 2NF, decompose the partial dependencies:

R1(A, B, C) for AB → C

R2(A, D, G, H) for AD → GH

R3(B, D, E, F) for BD → EF

R4(A, I) for A → I

We also need a relation containing the candidate key:

R5(A, B, D) for candidate key

Since H → J is a transitive dependency, if we also want to remove that redundancy and reach 3NF, create:

R6(H, J) for H → J

So, 6 tables are formed.

Q13. Relation R having attributes as (A, B, C, D, E) having the functional dependencies as follows:

A -> B
B -> E
C -> D

Identify that relation is in which normal form and the candidate key by determining the prime and non-prime attributes.
Apply the decomposition if needed to remove the redundancy.
Identify how many new tables will be formed after the decomposition.

Solution: Given, 1
→ 5
B → 9

C → D

A and C never appear on RHS, so A and C must be part of every candidate key.

Finding closures,
AC+ = ABCDE

Candidate Key: {AC}
Prime Attributes: {A, C}
Non-Prime Attributes: {B, D, E}

Checking BCNF,
A → B (A is not a super key, so it violates BCNF)

Checking 3NF,
A → B (A is not a super key and B is non-prime, so it violates 3NF)

Checking 2NF,
A → B (A is a proper subset of candidate key AC and B is non-prime, so it violates 2NF)

Highest NF: 1NF

Decomposition
To remove the partial and transitive dependencies, decompose as:

R1(A, B) for A → B

R2(B, E) for B → E

R3(C, D) for C → D

R4(A, C) for Candidate key AC

Number of new tables formed:

4 tables